Inventory Monitoring & Replenishment Policy

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1. Purpose and Operating Context

The Inventory Monitoring Service combines warehouse inventory, customer orders, supplier commitments, product master data, and recent demand signals. Its primary purpose is to identify products that may require replenishment before available stock falls below the operational safety threshold. The service does not place purchase orders automatically; instead, it creates a replenishment recommendation for an operations analyst to review.

A recommendation is considered actionable only when the product identifier is valid, the warehouse is active, inventory has passed its freshness check, and the projected available quantity is below the reorder point. Demand forecasts may be unavailable for new products, in which case the system falls back to the seven-day average daily demand.

1.1 Decision Flow

1. **Validate:** confirm SKU, warehouse, supplier, and inventory timestamps.
2. **Calculate:** estimate projected stock using on-hand quantity, reservations, inbound stock, and expected demand.
3. **Compare:** evaluate projected stock against the reorder point and safety stock.
4. **Prioritize:** assign urgency based on stockout risk, supplier lead time, and customer impact.

2. Inventory and Reorder Rules

The core calculation intentionally distinguishes physical stock from usable stock. On-hand quantity represents units physically recorded at the warehouse. Reserved quantity represents units already committed to confirmed customer orders. Inbound quantity represents supplier shipments that have been confirmed but have not yet been received.

Projected available = on-hand − reserved + confirmed inbound − expected demand during lead time.

The reorder point is calculated from expected demand during the supplier lead time plus the configured safety stock. A product should normally be flagged when projected available is less than the reorder point. However, an emergency flag is raised when projected available is negative or when the next confirmed customer shipment would create a stockout.

Table 1. Example replenishment candidates

SKU	Warehouse	On-hand	Reserved	Inbound	Reorder point	Status
SKU-1042	BLR-01	420	180	300	500	REPLENISH
SKU-2097	HYD-02	95	70	0	160	URGENT
SKU-3318	BLR-01	850	120	500	400	HEALTHY
SKU-4405	CHE-01	210	190	250	180	WATCH
SKU-5188	HYD-02	60	75	100	130	URGENT

The table is deliberately positioned between explanatory sections. During a retrieval experiment, questions about SKU-2097, SKU-4405, or the meaning of the Status column can reveal whether the parser preserves table structure and surrounding context.

3. Data Freshness and Exceptions

Inventory records are considered fresh when their source timestamp is no more than four hours old. Order reservations use a two-hour freshness threshold because customer-facing availability can change quickly. Supplier confirmations may be older when the shipment is still in transit, but the expected arrival date must remain valid.

If an inventory feed is stale, the service must not silently treat the last known quantity as current. Instead, it marks the warehouse feed as STALE and suppresses automatic replenishment recommendations for affected SKUs. Analysts may still inspect the records, but the dashboard should clearly display the timestamp and freshness state.

4. Supplier Lead Time and Priority

Supplier lead time is measured from confirmed purchase-order acceptance to expected receipt at the destination warehouse. The monitoring service uses supplier-specific lead times rather than a single global value because the same SKU can have different replenishment behavior depending on the supplier and warehouse combination.

4.1 Priority Classification

- **Critical:** projected available quantity is negative, or a confirmed customer order is expected to stock out before replenishment arrives.
- **High:** projected available quantity is below safety stock and supplier lead time exceeds three days.
- **Medium:** projected available quantity is below reorder point but remains above safety stock.
- **Low:** inventory is healthy but demand is increasing rapidly enough to require monitoring.

Priority is recalculated whenever a material inventory, order, or supplier event arrives. A recommendation generated at 09:00 can therefore move from Medium to Critical later the same day without any change to the product master record.

5. Example: SKU-2097 at HYD-02

SKU-2097 illustrates why a simple on-hand threshold is insufficient. The warehouse reports 95 units on hand, but 70 units are already reserved for confirmed orders. No inbound shipment is currently confirmed, while the reorder point is 160 units. The projected available quantity before considering future demand is therefore only 25 units.

The correct interpretation is not simply “95 units remain.” An analyst needs to understand that 70 units are committed, no replenishment is confirmed, and the reorder point is 160. Because the projected quantity is substantially below the reorder point and supplier lead time is four days, the item receives an URGENT recommendation.

5.1 Analyst Action

1. Check whether the reservation quantity is consistent with the order system.
2. Verify the last inventory update timestamp for HYD-02.
3. Review supplier availability and expected lead time.
4. Escalate to procurement if no inbound shipment can arrive before the projected stockout.

6. Monitoring Metrics

Metric	Definition	Target
Inventory freshness	Percentage of active warehouse feeds updated within 4 hours	≥ 98%
Recommendation latency	Time from source event to visible recommendation	< 15 min
Stockout precision	Percentage of Critical alerts confirmed as genuine risks	≥ 85%
Supplier coverage	Percentage of active SKUs with valid supplier lead time	≥ 95%

7. Auditability and Data Lineage

Every recommendation should be traceable to the source records used to calculate it. At minimum, the audit record stores SKU, warehouse_id, calculation timestamp, inventory timestamp, reservation timestamp, supplier identifier, lead time, reorder point, and the rule that produced the final priority. This makes it possible to distinguish a genuine inventory problem from a delayed or inconsistent source feed.

When records from multiple systems disagree, the monitoring service should preserve the individual source values rather than overwriting one system with another. For example, if the order system reports 70 reservations while the warehouse system reports 65, the discrepancy should be surfaced for reconciliation. The final recommendation must include enough metadata for an analyst to reproduce the calculation.

8. Benchmark Questions

The following questions are intentionally designed to test retrieval across headings, paragraphs, lists, and tables. They can be converted directly into evaluation queries with ground-truth answers for a Docling-versus-recursive-chunking benchmark.

1. What conditions make a replenishment recommendation actionable?
2. What is the projected available quantity formula?
3. Which SKU in the table has 70 reserved units and no inbound stock?
4. Why is SKU-2097 classified as URGENT?
5. What happens when an inventory feed is older than four hours?
6. What are the four priority levels and their conditions?
7. Which metrics have targets of at least 98% and at least 95%?
8. What information must be stored in the audit record?

Benchmark note: This document intentionally places related information in different representational forms. A recursive character splitter may separate a heading from its explanation, break a list across chunks, or fragment table context. Docling's structured document representation is designed to retain text, tables, hierarchy, reading order, and provenance. ■cite■turn0search8■turn0search0■